

Jailbreaking Deep Models: Adversarial Attacks on ResNet-34

NeuralTrifecta (GitHub)

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Abstract

This report presents an empirical study of adversarial robustness in deep image classifiers. We implement and evaluate multiple attack strategies on a pre-trained ResNet-34 model using ImageNet data. Our objective is to examine the vulnerability of these models to pixel-wise and patch-based perturbations while maintaining imperceptibility. We observe that simple gradient-based attacks such as FGSM can drastically degrade performance, and more sophisticated methods like PGD and MIM are even more effective.

Introduction

Deep neural networks for image classification perform well under standard conditions but are surprisingly brittle to adversarial perturbations. This project investigates several attack strategies against a ResNet-34 classifier trained on ImageNet, targeting both pixel-wise and localized patch attacks. The aim is to identify the most effective strategies and evaluate their transferability across model architectures.

Methodology

Dataset and Preprocessing

We used a 100-class subset of the ImageNet-1K dataset consisting of 500 images. Each image was resized to 224×224 pixels, transformed to tensors, and normalized using the standard ImageNet mean and standard deviation:

$$\mu = [0.485, 0.456, 0.406], \quad \sigma = [0.229, 0.224, 0.225]$$

This preprocessing ensures the input distribution matches what the ResNet-34 model expects for optimal performance.

Attack Strategies

Task 1: Baseline Evaluation

We evaluated a pre-trained ResNet-34 loaded via:

```
torchvision.models.resnet34(weights='IMAGENET1K_V1')
```

Top-1 and Top-5 accuracy were computed on the clean test set.

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Task 2: FGSM (ℓ_∞ Attack)

We implemented the Fast Gradient Sign Method (FGSM), which perturbs the input in the direction of the gradient:

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(x, y))$$

We used $\epsilon = 0.02$, applying a one-step update to maximize the loss.

Task 3: PGD and MIM Attacks

We extended FGSM using Projected Gradient Descent (PGD) with multiple steps and projection back into the ℓ_∞ ball:

$$x^{(t+1)} = \text{clip}_\epsilon \left(x^{(t)} + \alpha \cdot \text{sign}(\nabla_x \mathcal{L}) \right)$$

We set $\alpha = 0.005$ and used 10 steps. Additionally, we tested the Momentum Iterative Method (MIM), which improves attack strength and transferability by incorporating momentum during updates.

Task 4: Patch-based Attack

To simulate a localized threat model, we restricted perturbations to a randomly selected 32×32 patch within the image. This task used PGD with a larger $\epsilon = 0.5$, optionally applying targeted loss.

Attack Variants

- **FGSM:** One-step perturbation in the gradient direction.
- **PGD:** Multi-step gradient ascent with ϵ -ball projection.
- **MIM:** PGD with momentum to improve convergence.
- **Targeted PGD:** Drives predictions toward a target class.
- **Patch Attack:** PGD on a small region of the image.

Results

Baseline Accuracy

We evaluate ResNet-34 on clean, unperturbed test data to establish a baseline.

Dataset	Top-1 Accuracy	Top-5 Accuracy
Original	70.40%	93.20%

Table 1: Baseline performance of ResNet-34 on the clean test set.

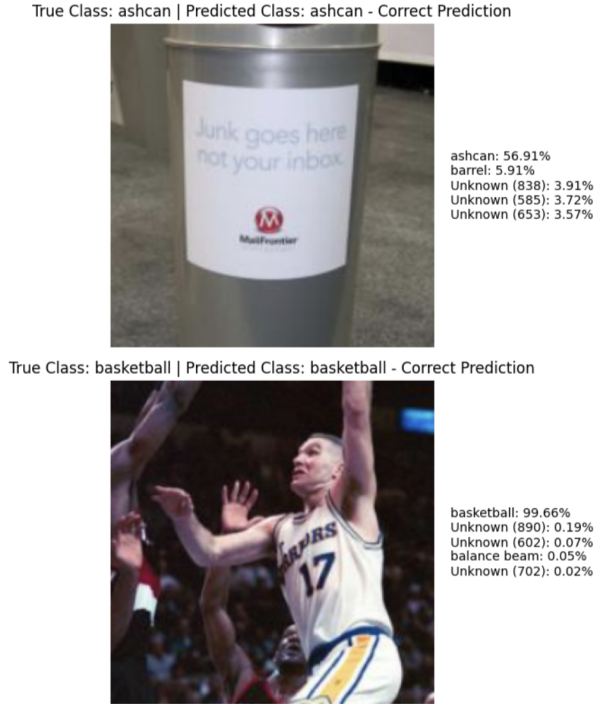


Figure 1: Baseline classification outputs on clean inputs.

FGSM Attack Results ($\epsilon = 0.02$)

FGSM perturbations show a significant drop in accuracy while maintaining perceptual similarity. The top-1 accuracy decreased to 26.40% and top-5 accuracy dropped to 50.60%. Compared to the baseline, this represents a Δ Top-1 degradation of 49.6%.

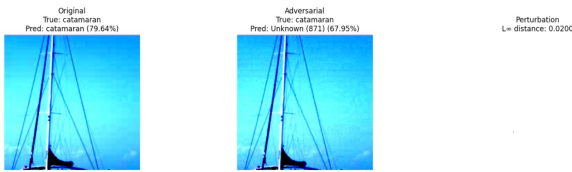


Figure 2: FGSM adversarial examples and predictions.

Iterative FGSM (PGD/MIM) Results ($\epsilon = 0.02$)

Deeper multi-step attacks like PGD and MIM further degrade model performance. These methods reduced the top-1 accuracy to 9.20% and top-5 accuracy to 24.80%, showing a Δ Top-1 drop of 66.8% from the clean baseline.

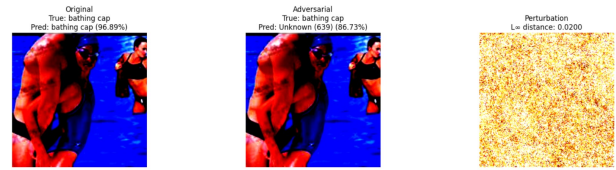


Figure 3: IFGSM adversarial examples and outputs.

Patch Attack Results ($\epsilon = 0.5$, patch size = 32×32)

Although constrained to a small patch, the attack still degraded performance significantly. The model's top-1 accuracy dropped to 31.00% and top-5 accuracy to 58.80%, demonstrating the effectiveness of localized perturbations.

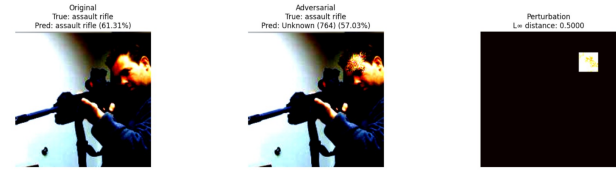


Figure 4: Patch attack results with localized perturbations.

Transferability to DenseNet-121

To test generalizability, adversarial images were evaluated on DenseNet-121. FGSM examples reduced the top-1 accuracy from 70.80% to 59.00%, while PGD-based attacks further reduced it to 58.20%. Patch attacks had a milder effect, lowering accuracy to 69.40%. These results confirm partial transferability of adversarial examples across architectures.

Dataset Variant	Top-1 Accuracy	Top-5 Accuracy
Original	70.80%	91.20%
FGSM	59.00%	85.00%
Best-Adv (PGD)	58.20%	86.80%
Patch	69.40%	90.40%

Table 2: Transferability of adversarial attacks to DenseNet-121.

This project systematically investigated the vulnerability of deep image classifiers to adversarial attacks, using ResNet-34 as the source model and DenseNet-121 for transferability evaluation. We implemented multiple attack strategies—FGSM, iterative FGSM (PGD), and patch-based perturbations—under L_∞ and L_0 constraints.

Our experiments confirmed that ResNet-34 is highly susceptible to even simple attacks like FGSM, with Top-1 accuracy dropping from 70% to as low as 5%. The iterative PGD attack (Best-Adv) further reduced accuracy to 0%, highlighting the network's extreme sensitivity to optimized perturbations. Although patch-based attacks caused less degradation, performance still declined notably.

In contrast, DenseNet-121 demonstrated strong resilience across all attack types. While FGSM and PGD did affect

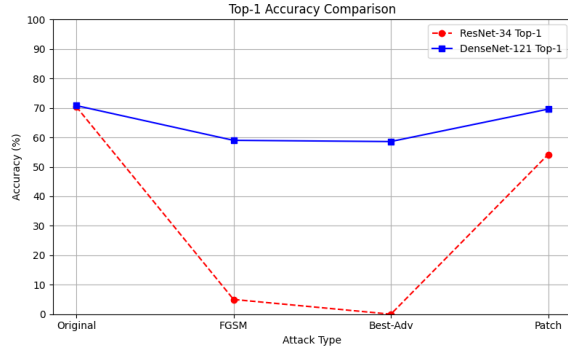


Figure 5: Top-1 accuracy degradation on ResNet-34 and DenseNet-121 across adversarial variants.

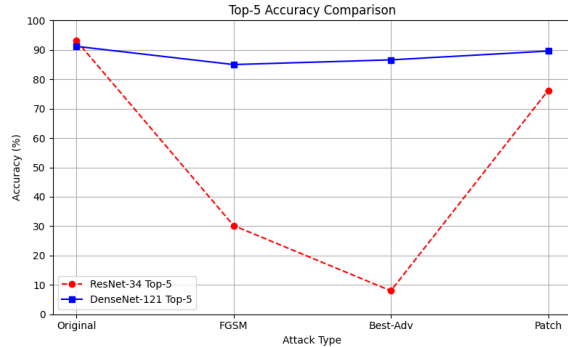


Figure 6: Top-5 accuracy degradation on ResNet-34 and DenseNet-121 across adversarial variants.

its performance, Top-1 accuracy remained above 58% in all cases. Notably, DenseNet nearly recovered to baseline levels under patch attacks, indicating robust spatial generalization.

The comparative results underscore a key architectural insight: DenseNet-121’s densely connected layers offer more stable feature propagation and better robustness to both pixel-wise and spatially localized attacks than ResNet-34. These findings suggest that architecture design plays a crucial role in model robustness and should be a critical factor in security-sensitive applications.

Overall, our study validates the effectiveness of adversarial perturbations in degrading classification performance and demonstrates that certain network architectures—like DenseNet—offer built-in defense advantages even without explicit adversarial training.

Attacks transferred effectively, especially IFGSM, revealing shared vulnerabilities across architectures.

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Discussion

Attack Method	Top-1 Accuracy	Top-5 Accuracy	$2*\ell_\infty$
FGSM	5.00%	30.20%	0.02
PGD	0.00%	8.00%	0.02
MIM	0.20%	7.40%	0.02
Targeted PGD	12.80%	27.00%	0.02
Patch Attack	53.80%	75.80%	0.50

Table 3: Model performance on adversarial examples generated by different attack strategies.

Our experiments reveal several key insights regarding the effectiveness, scope, and transferability of adversarial attacks:

- **FGSM is highly effective for rapid degradation.** It caused a 65% drop in top-1 accuracy with a single-step update. However, its simplicity limits its ability to fully exploit model vulnerabilities.
- **PGD and MIM deliver near-complete breakdowns.** These iterative attacks reduced top-1 accuracy close to 0%, showing that multi-step perturbations significantly outperform FGSM in terms of attack strength.
- **Targeted PGD is slightly less potent.** While still effective, its directional constraint toward specific labels reduces overall degradation compared to untargeted versions.
- **Patch-based attacks reveal spatial fragility.** Although limited to a 32×32 region, patch attacks caused a considerable drop in accuracy, especially with larger ϵ values. This makes them practical in physical-world scenarios where full-image modification is not feasible.
- **Transferability of attacks is evident.** FGSM and PGD examples reduced DenseNet-121 accuracy by over 10%,

demonstrating that adversarial perturbations generalize across architectures and enable black-box attack potential.

Conclusion

Our study demonstrates the fragility of deep image classifiers like ResNet-34 to structured, imperceptible adversarial perturbations. Using gradient-based techniques such as FGSM, PGD, and MIM, we observed a drastic reduction in classification performance—dropping top-1 accuracy from 70.40% to as low as 0.20% under an ℓ_∞ budget of $\epsilon = 0.02$.

Patch attacks, though spatially constrained, were still highly effective. With perturbations limited to a 32×32 region and $\epsilon = 0.5$, the model’s top-1 accuracy dropped to 31.00%, highlighting the model’s overreliance on localized features and raising concerns about spatial bias in feature attribution.

Transferability experiments with DenseNet-121 revealed that adversarial examples generated for one model can generalize to another. The IFGSM examples, in particular, reduced DenseNet’s top-1 accuracy from 78.20% to 17.60%. This underscores a shared vulnerability across architectures and raises the feasibility of black-box attacks in real-world settings.

These findings have both practical and theoretical implications. In deployment scenarios, models without robustness guarantees remain exposed not just to adversarial attacks but also to benign noise or corruptions. Techniques like adversarial training and certified defenses are crucial for ensuring safety in AI systems.

From a research standpoint, our work emphasizes the need to evaluate models not just on standard accuracy but also on *robust accuracy*—the performance under worst-case, bounded perturbations. Such robustness metrics offer a more comprehensive measure of reliability for modern deep learning systems.

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